

MagicCode Super-Resolution Software Release Notes

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1. Summary

MagicCode Super-Resolution Software v1.2.0 updates public mode names to SPEED and BALANCED, adds Direct3D 11 and CPU planar RGB input, introduces sharpen_level and combined parameter bins, and keeps the 1.0x–8.0x scaling range with CPU and GPU backends.

2. Key Changes

- Expanded scaling range to 1.0x through 8.0x, subject to backend and model support.
- Public algorithm modes are SPEED_MODE (0, formerly HIGH_SPEED_MODE) and BALANCED_MODE (1, formerly SPEED_MODE). Numeric values are unchanged.
- Runtime backend selection supports MAGIC_BACKEND_X86, MAGIC_BACKEND_NEON, MAGIC_BACKEND_METAL, MAGIC_BACKEND_OPENGL, MAGIC_BACKEND_OPENGL_ES, MAGIC_BACKEND_VULKAN, MAGIC_BACKEND_D3D11, and MAGIC_BACKEND_DEFAULT.
- Input types include CPU R8/planar RGB buffers and GPU textures: INPUT_BUFFER, INPUT_BUFFER_R8, INPUT_BUFFER_RGB, INPUT_TEXTURE_RGB8Unorm, and INPUT_TEXTURE_R8Unorm.
 - Unity integration includes Android and iOS wrapper support, C# bindings, and smoke validation workflow.
 - Unreal Engine integration includes Blueprint/C++ bridge APIs and Android/iOS-oriented plugin configuration.
 - Runtime status query exposes output dimensions, backend, input type, thread count, GPU time, and error code.
 - Technical reporting now includes additional runtime and performance fields for diagnostics, such as report reason and processing statistics.

3. Compatibility

- Operating systems: Windows 10 64-bit and above, Android 8.0 and above, iOS 13.0 and above, iPadOS 13.0 and above, and macOS 12.0 and above.
- CPU architectures: x86_64 and ARM64.
- Required SIMD instruction sets: AVX2 for x86_64 paths and NEON for ARM64 paths.
- Input resolution range: minimum 64 x 64, maximum 4032 x 4032.
- Maximum CPU thread count: 8. Multi-threading is used when input height is at least 256 rows.

4. Migration Notes

- Rename HIGH_SPEED_MODE to SPEED_MODE and SPEED_MODE to BALANCED_MODE. Prefer combined models magic_sr_gpu_params.bin and magic_sr_cpu_params.bin. Set sharpen_level in [0, 5].
 - Verify model files match the selected backend, mode, and scaling factor.
- Update integration code to populate input_param_t fields, especially input_type, backend, log_level, and sharpen_level.
 - Review the updated Privacy Policy and Data Reporting Notice before distributing applications that include the SDK or plugins.

5. Known Limitations

- Not every backend supports every input type, scaling factor, or model combination.
- CPU backends use buffer input (R8 or planar RGB); GPU backends use texture input.
- Performance depends on device hardware, graphics driver, backend, model, input resolution, scaling factor, and integration method.

- Unreal Engine iOS packaging may require a compatible Xcode and Apple SDK version for the selected UE release.

6. Validation Snapshot

- Unity Android and iOS smoke tests passed with v1.2.0, producing valid 128x128 output from 64x64 input.
- Unreal Engine Android smoke test passed with native path `MC_Init -> MC_Control(QUERY_STATUS) -> MC_Process -> MC_Uninit`.
- Unreal Engine iOS bridge configuration is included; local buildability depends on the installed UE/Xcode toolchain compatibility.